Free-Body Diagrams and Net Force

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

Which forces act on an object, and which way do they add up?

Targets

- I can name the two objects in any force interaction.
- I can sort forces as contact or field forces.
- I can draw a free-body diagram and find the net force.

Concepts

- **A force is an interaction** (F on A by B): A force is a push or pull by one object on another. Name both objects. An object cannot push itself.
- **The force catalog** (W = m g): Gravity (weight) pulls down. It is a field force. Normal force pushes away from a touching surface. Tension pulls along a rope. Friction opposes sliding. Applied force is a direct push or pull.
- **Free-body diagram** (arrows on ONE object): One object, one arrow per force on it. Arrow length shows the size of the force. Motion is not a force. Never draw one.
- **Net force** ($F_{net} = \Sigma F$): Add forces that point the same way. Subtract opposite ones. The net force points the way the object speeds up. Zero net force: still, or constant velocity.

What the class predicted, and what is true

- A book rests on a table. How many forces act on the book? **Two: gravity down and the table's push up.** Gravity pulls down and the table pushes up. They are equal, so the net force is zero.
- A crate slides to the right at constant speed. A person pushes it. How does the push compare with friction? **The push equals friction.** Constant speed means zero net force, so the push and friction are equal.
- The same crate now speeds up to the right. Which way does the net force point? **To the right.** The net force points the way the crate speeds up: the push beats friction.

Huddle notes (answers filled in)

Definitions

- **Free-body diagram:** A **picture** of one object with an **arrow** for each force on it.
- **Normal force:** The push of a **surface** on an object, **perpendicular** to the surface.
- **Net force:** The **sum** of all the forces on an object, with **direction**.

Worked example

A crate has W = 40 N down, N = 40 N up, a push of 20 N right, and friction of 8 N left. Find the net force.

- Up and down: 40 N and 40 N **cancel**.
- Right and left: 20 − 8 = **12** N.
- Net force: 12 N to the **right**.
- The crate **speeds up** to the right.

Label the crate

1. The **applied force** pushes right.
2. **Friction** opposes the slide.
3. **Gravity** pulls down.
4. The **normal force** pushes up.

One worked sample of each skill

Name the objects in an interaction

Level 1. A person pushes a wall. Which object exerts the push on the wall?
(A) The person (B) The wall (C) The floor (D) The air
Answer: A The person's hands touch the wall and push it.
Misconception targeted: Students let an object exert a force on itself, or name "motion" as the source.

Level 2. A book rests on a table. Which two objects interact through the normal force?
(A) The book and the table (B) The book and the floor (C) The book and the air (D) The table and the floor
Answer: A The normal force is a push between two surfaces that touch: the book and the table.
Misconception targeted: Students name one object, or a pair that does not touch.

Level 3. Which force would still act on a satellite-free rock in empty space near the Earth?
(A) Gravity (B) Normal force (C) Friction (D) Tension
Answer: A Gravity is a field force: the Earth pulls without touching. Normal force, tension and friction need contact.
Misconception targeted: Students call gravity a contact force because it "touches the ground".

Level 4. A banner hangs from a rope fixed to the ceiling. Which list names every object that exerts a force ON the banner?
(A) The rope and the Earth (B) The rope, the Earth and the ceiling (C) The rope only (D) The ceiling and the Earth
Answer: A The rope touches the banner and pulls up. The Earth pulls down at a distance. The ceiling touches the rope, not the banner.
Misconception targeted: Students include every object in the scene, not just those acting on this object.

Sort and name the forces

Level 1. A rough floor opposes a sliding box. This force is called...
(A) Friction (B) Normal force (C) Gravity (D) Tension
Answer: A A force that opposes sliding between two surfaces is friction.
Misconception targeted: Students mix up normal force and gravity because both are vertical.

Level 2. A book rests on a flat table. Which way does the normal force on the book point?
(A) Up (B) Down (C) Left (D) Right
Answer: A The normal force is perpendicular to the surface and pushes away from it: up.
Misconception targeted: Students think friction points the way the object moves, or that gravity stops at the top of a throw.

Level 3. Is applied force a contact force or a field force?
(A) Contact force (B) Field force (C) Neither
Answer: A Applied force needs the two objects to touch: a contact force.
Misconception targeted: Students call gravity a contact force, or call normal force a field force.

Level 4. A cart is pushed right across a rough floor. Which list names every force on it?
(A) W, N, F, f (B) W, N, F (C) W, N, F, f, force of motion (D) F, f
Answer: A Gravity (W) and the floor's normal force (N) act on it, plus the pull or push and friction from the rough floor. Nothing else touches it.
Misconception targeted: Students add a "force of motion", or forget the normal force and friction.

Pick the free-body diagram

Level 1. A phone lies on a shelf. Which diagram shows the forces on the phone?
(A) W down, N up; N equals W (B) W down (C) W down, N up, F right; N equals W (D) W down, N up; N shorter than W
Answer: A Gravity pulls down and the surface pulls or pushes up with an equal arrow, so the net force is zero.
Misconception targeted: Students draw only gravity on a still object, or add a force for "staying there".

Level 2. A crate slides left at constant speed. A person pushes it left. Which diagram is right?
(A) W down, N up, F left, f right; N equals W; F equals f (B) W down, N up, F left; N equals W (C) W down, N up, F left, f right, M left; N equals W; F equals f (D) W down, N up, F left, f right; N equals W; F longer than f
Answer: A Constant speed means net force zero: push equals friction, and N equals W.
Misconception targeted: Students believe motion needs a bigger forward force than friction, or draw a "force of motion".

Level 3. A trunk speeds up to the right. A person pushes it right. Which diagram is right?
(A) W down, N up, F right, f left; N equals W; F longer than f (B) W down, N up, F right, f left; N equals W; F equals f (C) W down, N up, F right, f left, M right; N equals W; F longer than f (D) W down, N up, F right; N equals W
Answer: A Speeding up means the net force points right: the push arrow is longer than friction. N still equals W.
Misconception targeted: Students draw push and friction equal for any motion.

Level 4. A person pushes a crate right with 35 N. Friction is 10 N and the crate weighs 40 N. It speeds up. Which diagram has arrows drawn to scale?
(A) W down (40 N), N up (40 N), F right (35 N), f left (10 N); N equals W; F longer than f (B) W down (40 N), N up (30 N), F right (35 N), f left (10 N); N shorter than W; F longer than f

(C) W down (40 N), N up (40 N), F right (10 N), f left (35 N); N equals W; F shorter than f (D) W down (40 N), N up (40 N), F right (35 N); N equals W
Answer: A N equals W (40 N). The right arrow (35 N) is longer than the left (10 N), so the net force is 25 N to the right.
Misconception targeted: Students draw lengths that do not match the numbers, or let N differ from W.

Which way is the net force?

Level 1. A box has 6 N pushing right and 6 N pushing left. Which way is the net force?
(A) It is zero (B) Right (C) Left
Answer: A Equal arrows in opposite directions cancel: the net force is zero.
Misconception targeted: Students think equal forces still "push" the box one way.

Level 2. A crate has 9 N pulling left and 7 N of friction pulling right. Which way is the net force?
(A) Left (B) Right (C) It is zero
Answer: A The bigger force is 9 N, left. The net force is 2 N left.
Misconception targeted: Students take the net force along the motion, or call friction the "smaller force" and ignore it.

Level 3. A crate weighs 60 N, the floor pushes up 60 N, a person pushes right 8 N, and friction is 4 N left. Which way is the net force?
(A) Right (B) It is zero (C) Left (D) Up
Answer: A Up and down cancel (60 N each). Right 8 N and left 4 N: net 4 N right.
Misconception targeted: Students try to add vertical and horizontal forces into one number.

Level 4. A 3 kg lamp (weight 29.4 N) hangs from a chain pulling up 18 N. Which way is the net force?
(A) Down (B) Up (C) It is zero
Answer: A $W = 3 \times 9.8 = 29.4$ N. The chain pulls 18 N. The weight is bigger, so the net force points down.
Misconception targeted: Students assume a hanging object is always balanced.

Find the net force

Level 1. A cart is pushed right with 9 N and friction pulls left with 4 N. How big is the net force?
Answer: 5 N $9 - 4 = 5$ N, to the right.
Misconception targeted: Students add forces that point opposite ways.

Level 2. A crate is pulled right with 12.5 N. Friction is 8.5 N left. How big is the net force?
Answer: 4 N $12.5 - 8.5 = 4$ N.
Misconception targeted: Students drop the decimal or add the forces.

Level 3. A 5 kg box (weight 49 N, floor pushes up 49 N) is pushed right with 15 N. Friction is 10 N. What is the net force?
Answer: 5 N Up and down cancel. Net $= 15 - 10 = 5$ N, to the right.
Misconception targeted: Students include weight in a horizontal net force.

Level 4. Two people push a crate right with 23 N and 14 N while friction pulls left with 15 N. What is the net force?
Answer: 22 N Right: $23 + 14 = 37$ N. Left: 15 N. Net $= 37 - 15 = 22$ N right.
Misconception targeted: Students subtract in the order they read the numbers.

Two objects, one diagram

Level 1. A cup sits on a desk. Which force does NOT act ON the cup?
(A) The cup's push on the desk (B) Gravity on the cup (C) The desk's push up on the cup
Answer: A The cup pushes on the desk, so that force acts on the desk. A diagram of the cup shows only forces on the cup.
Misconception targeted: Students draw the object's force on the table as a force on the object.

Level 2. A book A sits on a book B, and B sits on a table. Which list names the forces on book A?
(A) Gravity on A, and B's push up (B) Gravity on A, B's push up, and the table's push up (C) Gravity on A and on B (D) Gravity on A only
Answer: A A is touched only by B, so B pushes up on it. The table touches B, not A.
Misconception targeted: Students list every force in the picture on every object.

Level 3. Box A (4 kg) sits on box B (1 kg). The table pushes up on B. How hard does the table push, in newtons? ($g = 9.8$ N/kg)
Answer: 49 N The table holds up both: $N = (4 + 1) \times 9.8 = 49$ N.
Misconception targeted: Students count only the lower box's weight.

Level 4. Boxes of 2 kg and 3 kg are tied together and pulled right with 34 N. Friction on the pair totals 12 N. What is the net force on the pair?
Answer: 22 N $34 - 12 = 22$ N to the right. The tension between the boxes is inside the system and does not count.
Misconception targeted: Students add the tension between the boxes into the net force.